

Data Flow Diagram

Date	14 July 2026
Team ID	
Project Name	HDI Predictor (Human Development Index Predictor)
Team Size	1 (Individual Project)
Maximum Marks	2 Marks

DFD Symbol Legend

Symbol	Name	Description
Oval	External Entity	User (browser) who submits indicators and views results.
Numbered rectangle	Process	Validate Input, Predict HDI, Classify Tier, Log Prediction.
Solid rectangle	Data Store	country_data.json, model.pkl, predictions.db (SQLite).
Labeled arrow	Data Flow	Indicator values, JSON prediction result, saved history row.

Level-1 Data Flow

1. **User** → submits life expectancy, schooling (expected & mean), and GNI via the Predict form.
2. **Process: Validate Input** → checks numeric ranges (e.g. life expectancy 0–100); invalid input returns an error to the User without touching the model.
3. **Process: Predict HDI** → loads **model.pkl** (Data Store), log-transforms GNI, runs the regression, and outputs a raw HDI score.
4. **Process: Classify Tier** → maps the HDI score to a UNDP tier using fixed thresholds.
5. **Process: Log Prediction** → writes the input + result to **predictions.db** (Data Store) with a timestamp.
6. **Predict HDI** and **Classify Tier** results flow back to the **User** as JSON, rendered as the gauge and tier badge.
7. On the History page, **User** requests logged data → **Process: Fetch History** reads from **predictions.db** and returns it for display.